

Discrete Compatibility Conditions

Analytical/Continuous

By divergence theorem

$$\int_V \nabla^2 \phi = \int_V \nabla \cdot (\nabla \phi) = \int_{\partial V} \nabla \phi$$

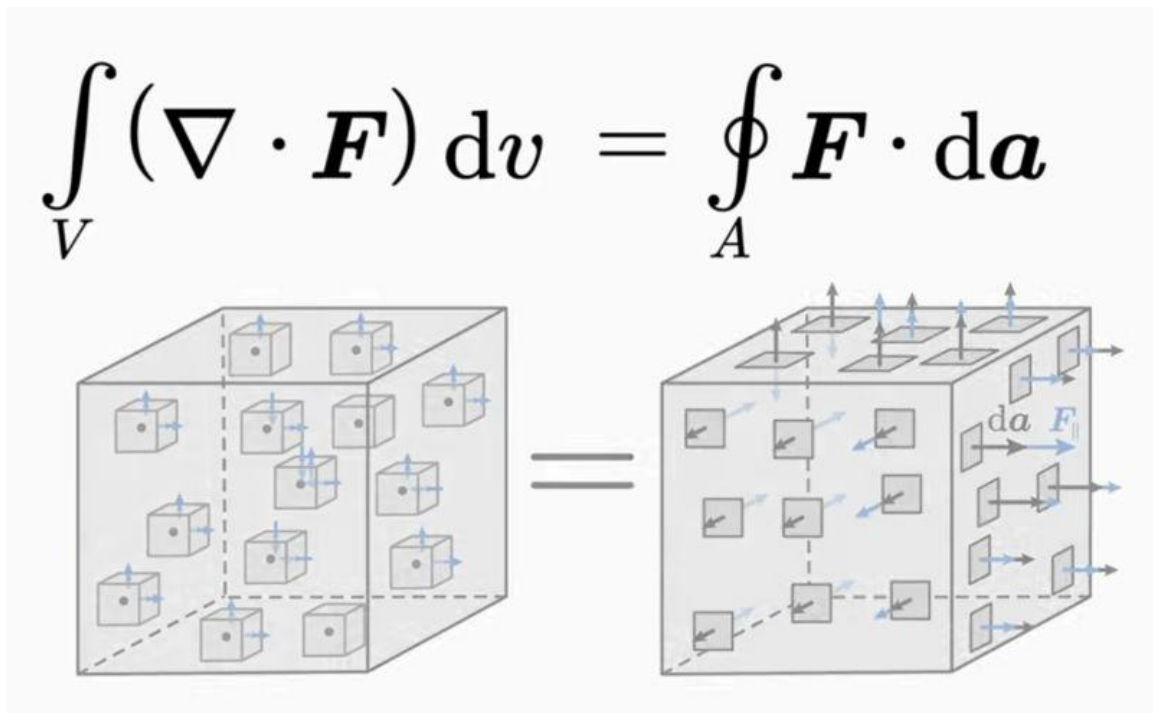
where ∂V denotes the boundary of V in the integration of the flux.

If the Neumann BC is

$$\nabla \phi(\partial V) = g,$$

then

$$\int_V \nabla^2 \phi = \int_{\partial V} g.$$



Numerical/Discrete

As shown in the figure above, the divergence theorem holds due to cancellation of the fluxes on neighboring interior boundaries, resulting in only the outermost boundary surviving the cancellation.

Hence we expect that in the discrete version, it should be due to some Telescoping Summations.

Assume rectangular grid and define

$\sum_V$: Summation over entire grid V .

$\sum_{\text{Int}(V)}$: Summation over interior points of V .

$\sum_{\partial V}$: Summation over the boundary points of V (excluding the corners).

Consider the discrete version of the integral of the Laplacian over $\text{Int}(V)$.

$$\sum_{\text{Int}(V)} (\phi_{j+1,l} + \phi_{j-1,l} + \phi_{j,l+1} + \phi_{j,l-1} - 4\phi_{j,l}).$$

All $\phi_{j,l}$ in $\text{Int}(\text{Int}(V))$ will be cancelled because each of them appears 5 times in the sum, once with coefficient -4 and four times each with coefficient 1 .

Points on $\partial(\text{Int}(V))$ each appears 4 times, once with coefficient -4 and three times with coefficient 1 , and hence survives with a coefficient -1 . (Except the 4 corner points of $\partial(\text{Int}(V))$ each surviving with coefficient -2 .)

Points on $\partial(V)$ each appears once with coefficient 1 . (Except the 4 corner points of V which never appear.)

Therefore, it is easy to check that

$$\sum_{\text{Int}(V)} (\phi_{j+1,l} + \phi_{j-1,l} + \phi_{j,l+1} + \phi_{j,l-1} - 4\phi_{j,l}) = \sum_{\partial V} (\phi_{\partial} - \phi_I), \quad (\text{Eq. 1})$$

where ϕ_∂ is the potential at a point on the boundary and ϕ_I is the potential of the interior point right next to it in the direction normal to the boundary.

By Poisson's equation, we have

$$\sum_{\partial V} (\phi_\partial - \phi_I) = \sum_{\text{Int}(V)} \rho_{ij} \Delta^2. \quad (\text{Eq. 2})$$

1. One-sided differencing for Neumann BC

Require that

$$\phi_\partial - \phi_I = g\Delta.$$

Then compatibility condition is

$$\sum_{\partial V} g\Delta = \sum_{\text{Int}(V)} \rho_{ij} \Delta^2.$$

2. Central differencing for Neumann BC (Preferred due to higher accuracy)

Need to set ghost grid points (denoted by G) just outside the boundary.

Require that

$$\phi_G - \phi_I = 2g\Delta$$

Apply Eq. (1) on the extended grid including the ghost layer,

$$\sum_V (\phi_{j+1,l} + \phi_{j-1,l} + \phi_{j,l+1} + \phi_{j,l-1} - 4\phi_{j,l}) = \sum_{\partial V} (\phi_G - \phi_\partial) = \sum_{\partial V} (\phi_I - \phi_\partial + 2g\Delta)$$

$$\sum_{\partial V} (\phi_\partial - \phi_I) = - \sum_V \rho_{ij} \Delta^2 + \sum_{\partial V} 2g\Delta$$

But by Eq. (2), we have

$$\sum_{\text{Int}(V)} \rho_{ij} \Delta^2 = - \sum_V \rho_{ij} \Delta^2 + \sum_{\partial V} 2g\Delta$$

Hence

$$\sum_{\partial V} g\Delta = \frac{1}{2} \left(\sum_{\text{Int}(V)} \rho_{ij}\Delta^2 + \sum_V \rho_{ij}\Delta^2 \right) = \sum_{\partial V} \frac{1}{2} \rho_{ij}\Delta^2 + \sum_{\text{Int}(V)} \rho_{ij}\Delta^2.$$